
The Geometry of Untranslatability: Generation Faithfulness and Language Identity in Multilingual LLMs

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Abstract

Prior work shows that multilingual LLMs often process meaning in an English-centric latent space, but almost exclusively for concepts that *have* English equivalents. We study the complementary case: culturally specific concepts with no single-word English translation. Across four open-weight models (Llama-3-8B, Llama-3-8B-Instruct, Mistral-7B-v0.3, Qwen2.5-7B) and a cleaned inventory of 90 untranslatable concepts with 122 matched controls (10 languages; 636 prompts), we obtain a three-part result. **(1) Behavioral.** Greedy generations scored with a reproducible faithfulness rubric are significantly worse for untranslatable concepts in all four models (Cohen’s $d \in [-0.56, -0.39]$, all $p < 10^{-4}$). **(2) Representational.** Linear probes with concept-grouped cross-validation recover source language nearly perfectly, while translatability is weaker but above chance—including within language. **(3) Cautionary.** English Routing Rate (ERR) shows *no* consistent “untranslatable→English” cost at the last token; early-layer peaks and tokenizer size dominate. Lexical gaps therefore leave a clearer fingerprint in *how models explain culture* and *where language identity lives* than in aggregate English-routing statistics.

1 Introduction

Multilingual large language models (LLMs) are used worldwide, yet mechanistic analyses of their internals remain dominated by an English lens. Recent work argues that key semantic decisions occur in a representation space closest to English [5, 6], and that romanization can bridge concept space and language-specific output [3]. These studies, however, largely examine content that *does* have English equivalents. A culturally consequential question remains open: what happens when a concept has *no* single-word English equivalent?

Consider Japanese *Tsundoku*—the habit of acquiring books and letting them pile up unread. When a model processes such a concept, three outcomes are plausible: (i) it forces an English approximation in the residual stream; (ii) it carves out an isolated, language-specific region; or (iii) its *explanations* of the concept degrade even when residual-stream English statistics look normal.

We test these hypotheses with three complementary measurements on a frozen, audited concept inventory:

1. **Generation faithfulness** (primary behavioral outcome): do models produce weaker cultural explanations for untranslatable concepts than for matched controls?
2. **Linear probes** (primary representational outcome): is *translatability* linearly recoverable from activations, or primarily *language identity*?

Model	Hugging Face ID	L	d	$ \mathcal{V} $
Llama-3-8B	Meta-Llama-3-8B	32	4096	128K
Llama-3-8B-Instruct	Meta-Llama-3-8B-Instruct	32	4096	128K
Mistral-7B	Mistral-7B-v0.3	32	4096	33K
Qwen2.5-7B	Qwen2.5-7B	28	3584	152K

Table 1: Model configurations (L : layers; d : residual width; $|\mathcal{V}|$: vocabulary size).

3. **English routing** (cautionary secondary): does the logit lens show elevated English probability for lexical gaps?

Contributions. (i) We release a cleaned untranslatability probe set with matching diagnostics and an exclusion audit. (ii) We show a consistent behavioral cost of lexical gaps under generation faithfulness across four models. (iii) We show that residual-stream geometry encodes language strongly and translatability more weakly, while ERR—a popular English-routing summary—does not detect the gap. Together, these results reframe how cultural lexical gaps should be measured in LLMs.

2 Related Work

English-centric latent spaces. Schut et al. [5] and Wendler et al. [6] provide evidence that multilingual transformers route semantically loaded content through English-adjacent representations. RomanLens [3] shows that intermediate layers often prefer romanized forms before resolving to native script. We extend this line of work to the lexical-gap regime and argue that ERR alone is a brittle primary metric for cultural claims.

Cultural representations. Sparse-feature analyses of cultural [7, 2] and language-specific [1] structure suggest that LLMs encode culture and language as partially separable features. We ask a sharper question: whether *absent* English equivalents—not just culture-tagged concepts—change generation quality or linear geometry.

Untranslatability. Lexical gaps are classical objects in linguistic relativity [4]. We operationalize them with a generation rubric, concept-grouped linear probes, and corrected logit-lens diagnostics.

3 Methodology

3.1 Dataset

We start from 227 culturally specific concepts across 10 languages (Arabic, Danish, German, Spanish, French, Hindi, Japanese, Korean, Portuguese, Russian) and apply a frozen audit that drops clearly invalid “untranslatable” labels (e.g., Greek *Kairos* mislabeled as Japanese; near-direct glosses such as Hindi *Dill* “heart”). The cleaned set contains **90 untranslatable** and **122 translatable** controls (212 concepts). Matching diagnostics record language $\times$ category coverage, character-length summaries, English-loanword flags, and greedy same-language control pairs.

Each concept is rendered with three prompt templates—definition, usage, and comparison—yielding 636 prompts. Generation experiments use definition and comparison (424 generations per model).

3.2 Models and extraction

We study four open-weight models (Table 1). For every prompt we record residual-stream activations at (i) the last non-padding token and (ii) the first token of the concept word, together with LM-head weights and the final RMSNorm gain required for a correct logit lens.

68 3.3 Generation faithfulness

69 For each concept we generate a greedy continuation ($T=0$, $\text{max_new_tokens}=128$) for definition
70 and comparison prompts. Let y be the generation, m the gold English gloss, w the concept word,
71 ℓ the source language, and $c \in \{\text{untrans}, \text{transl}\}$ the label. Writing $T(\cdot)$ for the set of alphabetic
72 tokens (lowercased) and $n=|y|$ for the whitespace token count, meaning overlap is

$$s_{\text{mean}}(y, m) = \min(1, 1.5 \cdot |T(y) \cap T(m)| / |T(m)|). \quad (1)$$

73 Cultural specificity accumulates additive rewards (clipped to $[0, 1]$):

$$s_{\text{spec}} = \text{clip}_{[0,1]}(0.25 \cdot \mathbf{1}[w \subseteq y] + 0.15 \cdot \mathbf{1}[\ell \subseteq y] \\ + 0.30 \cdot \mathbf{1}[n \geq 12] + 0.15 \cdot \mathbf{1}[n \geq 25] + 0.15 \cdot \mathbf{1}[\text{punct. in } y]), \quad (2)$$

74 and English-collapse penalties accumulate as

$$\tilde{s}_{\text{coll}} = \text{clip}_{[0,1]}(0.5 \cdot \mathbf{1}[n \leq 4] + 0.2 \cdot \mathbf{1}[n \leq 8] + 0.3 \cdot \mathbf{1}[\text{gloss pattern}]), \\ s_{\text{coll}} = \begin{cases} 0.5 \tilde{s}_{\text{coll}} & \text{if } c = \text{transl}, \\ \tilde{s}_{\text{coll}} & \text{otherwise,} \end{cases} \quad (3)$$

75 where ‘‘gloss pattern’’ matches short templates such as ‘‘means X ’’ / ‘‘is X ’’ / ‘‘refers to X ’’. The
76 composite faithfulness score is

$$s(y) = \text{clip}_{[0,1]}(0.45 s_{\text{mean}} + 0.40 s_{\text{spec}} - 0.25 s_{\text{coll}} + 0.20). \quad (4)$$

77 We compare untranslatable vs. translatable scores with Welch’s t -test and Cohen’s d . The rubric is
78 fully deterministic for reproducibility; an LLM-as-judge interface is provided in code as a drop-in
79 replacement.

80 3.4 Linear probes

81 At layer ℓ and token position $p \in \{\text{last}, \text{concept}\}$ we fit ℓ_2 -regularized logistic regression probes
82 to predict language identity or the binary translatability label. To prevent leakage across the three
83 templates that share a concept, we use **GroupKFold** with groups defined by (language, word).
84 We additionally fit within-language translatability probes that remove the language confound. Let
85 $\text{Acc}_{\text{lang}}(\ell)$ and $\text{Acc}_{\text{tr}}(\ell)$ denote mean CV accuracy; the quantity of interest is the gap

$$\Delta(\ell) = \text{Acc}_{\text{lang}}(\ell) - \text{Acc}_{\text{tr}}(\ell). \quad (5)$$

86 3.5 English routing (secondary)

87 The corrected logit lens projects a hidden state h_ℓ through final RMSNorm and the LM head:

$$z_\ell = W_{\text{U}} \text{RMSNorm}(h_\ell), \quad p_\ell = \text{softmax}(z_\ell). \quad (6)$$

88 Top-1 English Routing Rate at layer ℓ is

$$\text{ERR}_\ell = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{tok}(\arg \max p_\ell^{(i)}) \in V_{\text{en}}], \quad (7)$$

89 and the graded English probability mass is

$$p_{\text{eng}}(h_\ell) = \sum_{t \in V_{\text{en}}} p_\ell(t), \quad (8)$$

90 where V_{en} is the set of lowercase Latin-letter vocabulary tokens (after stripping BPE spacing markers).
91 Omitting final RMSNorm invalidates the projection for some architectures (notably Qwen); we
92 therefore always apply it.

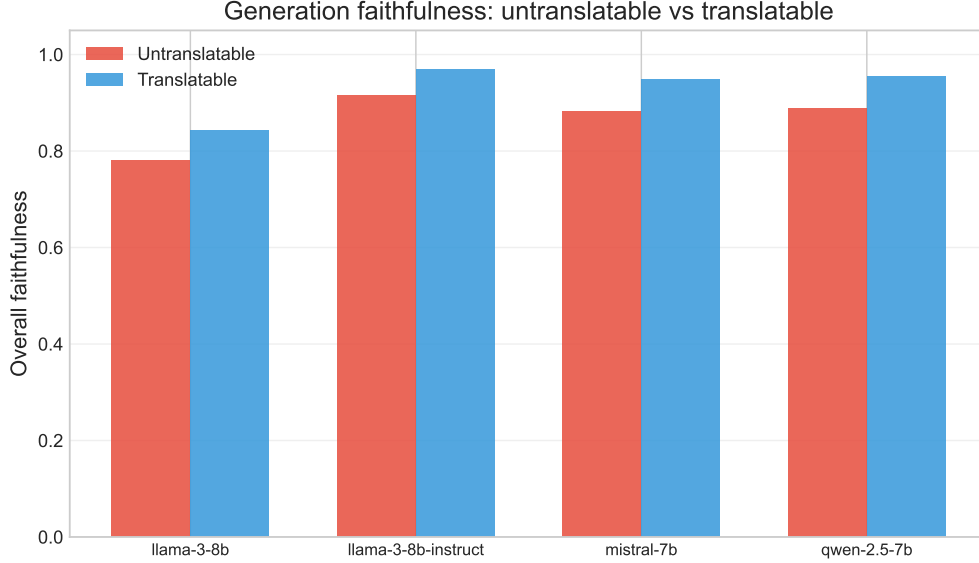


Figure 1: Overall generation faithfulness by model and group (higher is better).

Model	Mean untrans.	Mean transl.	Cohen’s d
Llama-3-8B	0.779	0.843	−0.39
Llama-3-8B-Instruct	0.916	0.968	−0.56
Mistral-7B	0.883	0.948	−0.55
Qwen2.5-7B	0.889	0.954	−0.55

Table 2: Generation faithfulness (Eq. 4). Negative d means worse scores for untranslatable concepts; all four effects are significant at $p < 10^{-4}$.

93 4 Results

94 4.1 Generation: a behavioral cost of lexical gaps

95 Figure 1 and Table 2 summarize $s(y)$ on 424 generations per model. **In all four models, untrans-**
 96 **latable concepts receive significantly lower faithfulness** ($d \in [-0.56, -0.39]$, all $p < 10^{-4}$). The
 97 effect is strongest on definition prompts ($d \in [-0.85, -0.67]$, all $p < 10^{-5}$); comparison prompts
 98 show smaller gaps that remain significant in three of four models. This is precisely the behavioral
 99 signature that last-token ERR does not reveal: models explain lexical-gap concepts less faithfully
 100 than matched controls.

101 4.2 Probes: language dominates, translatability is weaker

102 GroupKFold probes recover source language nearly perfectly at both positions (peak $\text{Acc}_{\text{lang}} \approx 1.0$;
 103 chance 0.1). Translatability accuracy is high but strictly lower (peak $\text{Acc}_{\text{tr}} \approx 0.94\text{--}0.96$), and
 104 within-language peaks remain above chance ($\approx 0.84\text{--}0.94$). Thus $\Delta(\ell) > 0$ consistently: language
 105 identity is the dominant linear structure, yet a residual lexical-gap direction exists that ERR fails to
 106 surface. Concept-token language ARI (Figure 3) likewise rises far above last-token ARI in late layers
 107 (peaks up to ~ 0.85 for Llama-3-8B-Instruct).

108 4.3 English routing: a null secondary finding

109 ERR is highest in early layers—where top-1 predictions reflect embedding/unembedding ge-
 110 ometry and high-frequency English tokens—and collapses at the output, where models pre-
 111 dict source-language continuations (Figure 4). Layer-wise Welch tests find *no* significant
 112 untranslatable > translatable ERR difference in any model (Table 3); trends often point the other way.

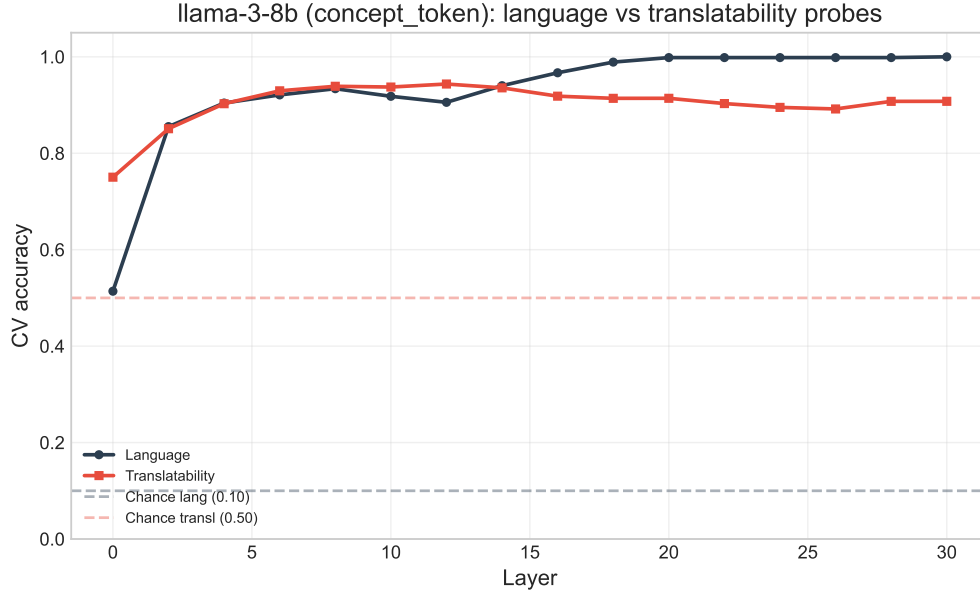


Figure 2: Concept-token probe accuracy for language vs. translatability (Llama-3-8B; GroupKFold by concept). Chance is 0.1 for language and 0.5 for translatability.

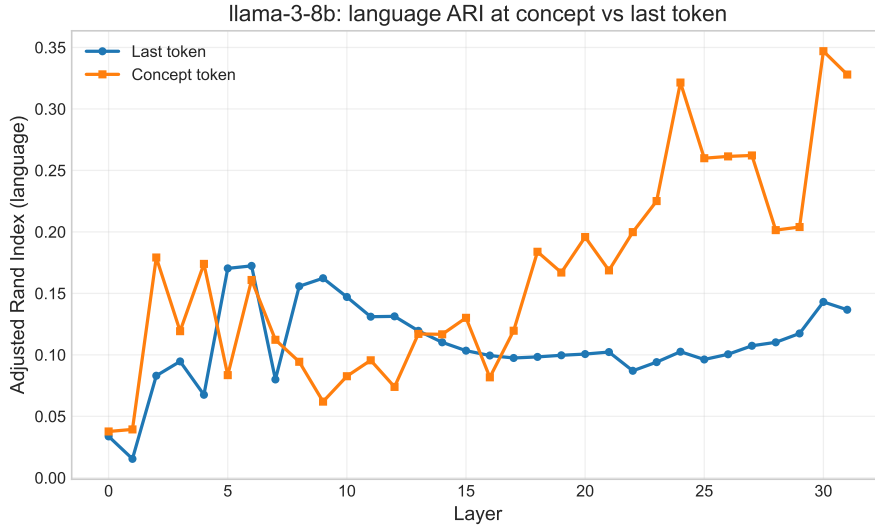


Figure 3: Language ARI across layers at last-token vs. concept-token positions (Llama-3-8B).

113 Graded mid-layer p_{eng} is likewise null in three of four models at the last token. Concept-token p_{eng}
 114 effects that do reach significance flip sign across architectures (Llama-3-8B $d = +0.27$; Qwen2.5-7B
 115 $d = -0.45$), so they cannot support a unified English-routing account of untranslatability.

116 Cross-model ERR correlations are high within Llama/Mistral ($r \approx 0.56$ – 0.72) but near zero or
 117 negative vs. Qwen, tracking tokenizer vocabulary structure rather than shared cultural semantics.
 118 Sparse dictionary features ($k=512$) likewise fail to separate groups better than chance at either token
 119 position.

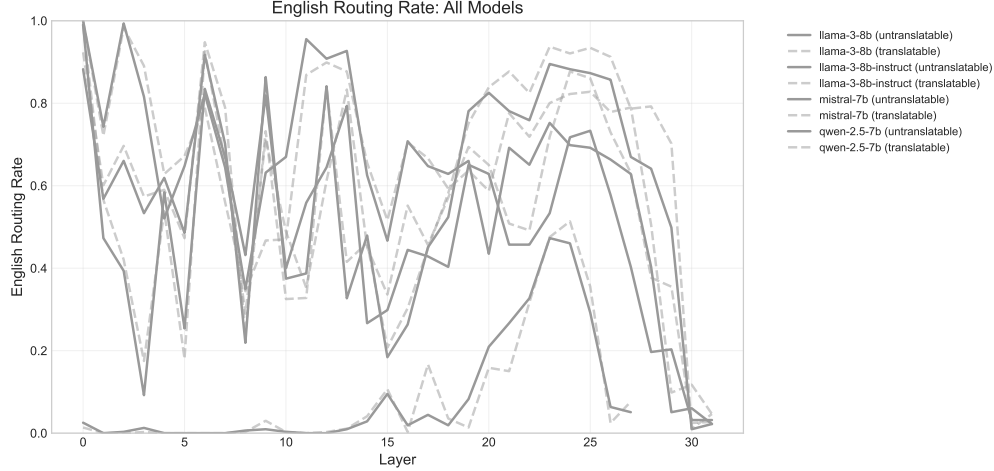


Figure 4: English Routing Rate (Eq. 7) across layers (untranslatable solid, translatable dashed).

Model	ERR untrans.	ERR transl.	Cohen’s d	p
Llama-3-8B	0.587	0.619	−0.12	0.629
Llama-3-8B-Instruct	0.481	0.533	−0.21	0.393
Mistral-7B	0.629	0.650	−0.09	0.721
Qwen2.5-7B	0.089	0.089	0.00	0.999

Table 3: Welch tests over layer-wise ERR (secondary analysis).

5 Discussion

Measure what culture does, not only where English peaks. Generation faithfulness and concept-token probes speak more directly to cultural representation than aggregate ERR. The null ERR result is informative: it warns against reading early-layer English mass as evidence of cultural bias.

Language identity dominates residual geometry. Near-perfect language probes and high concept-token ARI show that models encode *which language a word belongs to*. Within-language translatability probes above chance further suggest a residual lexical-gap correlate that does not appear as elevated English routing—a dissociation between probe geometry and logit-lens ERR.

Tokenizer and early-layer effects dominate routing summaries. Practical recommendations follow: always apply final RMSNorm in logit-lens analyses; discard early and output layers when summarizing routing; report tokenizer diagnostics alongside ERR; and prefer graded and behavioral metrics.

Limitations. The cleaned inventory is still modest relative to conversational cultural use. The faithfulness score in Eq. 4 is a reproducible lexical/structural proxy—not a human or LLM judge—and may under-penalize fluent but culturally shallow text; LLM-as-judge and human spot-checks are natural extensions. Single-concept prompts omit dialogue context. We do not run causal interventions (activation patching or steering).

Broader impacts. Positive: clearer metrics can reduce over-claiming of “English bias” from routing artifacts and encourage culturally faithful evaluation. Negative: lexical-gap inventories can encode annotator subjectivity and Western-centric notions of what counts as “untranslatable”; generation evaluations could be misused to rank languages by stereotyped “difficulty.” We mitigate by releasing the exclusion audit, matching diagnostics, and deterministic rubric so assumptions are inspectable.

142 **Compute.** Activation extraction and generation used NVIDIA L4 GPUs via Modal; full Path-B
143 analysis for four models required on the order of a few GPU-hours. Code, cleaned data, probe results,
144 and generations are released with the paper repository.

145 6 Conclusion

146 Untranslatable cultural concepts do not leave a reliable last-token English-routing signature, yet they
147 are not invisible: language identity is strongly recoverable at the concept token, within-language
148 probes detect a weaker lexical-gap direction, and generated explanations are consistently less faithful
149 for untranslatable concepts than for matched controls across four models. Measuring generation
150 quality and concept-token geometry therefore complements—and corrects—over-reliance on ERR
151 alone.

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